2.1 Unit description

Chemical ideas

This unit develops the treatment of chemical bonding by introducing intermediate types of bonding and by exploring the nature and effects of intermolecular forces.

Study of the periodic table is extended to cover the chemistry of groups 2 and 7. Ideas about redox reactions are applied in particular to the reactions of halogens and their compounds.

The unit develops a largely qualitative understanding of the ways in which chemists can control the rate, direction and extent of chemical change.

Organic chemistry in this unit covers alcohols and halogenoalkanes. The treatment is extended to explore the mechanisms of selected examples.

Students have to use formulae and balance equations and have an understanding of chemical quantities.

How chemists work

Electron-pair repulsion theory shows how chemists can make generalisations and use them to make predictions.

Chemists rationalise a great deal of information about chemical changes by using theory to categorise reagents and types of chemical change. This is illustrated by the use of inorganic and organic examples in this unit.

The use of models in chemistry is illustrated by the way in which the Maxwell-Boltzmann distribution and collision theory can account for the effects of temperature on the rates of chemical reactions.

The unit shows how chemists can study chemical changes on an atomic scale and propose mechanisms to account for their observations.

Chemistry in action

This unit shows the contribution that chemistry can make to a more sustainable economy by redeveloping manufacturing processes to make them more efficient, less hazardous and less polluting.

Insight into the mechanisms of chemical reactions can help to account for the damaging effects of some chemicals on the natural environment.

The study of spectroscopy gives further examples of the importance of accurate and sensitive methods of analysis which can be applied to study chemical changes but also to detect drugs such as alcohol.

The unit deals with issues regarding the environment, such as climate change, the effect of greenhouse gases, carbon footprints and other aspects of green chemistry. It ensures that students understand the underlying chemistry and can investigate ways to combat these issues.

Core practicals

The following specification points are core practicals within this unit that students should complete:

2.4d

2.5c

2.7.1g

2.7.2b

2.7.2c

2.7.2d

2.8f

2.10.1d

2.10.2c

2.10.2e

These practicals may appear in the written examination for Unit 2.

Use of examples

Examples in practicals

Where 'e.g.' follows a type of experiment in the specification students are not expected to have carried out that specific experiment. However, they should be able to use data from that or similar experiments.

For instance in this unit, *2.7g ii Properties down group 2*, the specification states:

simple acid-base titrations using a range of indicators, acids and alkalis, to calculate solution concentrations in g dm^{-3} *and mol dm*$^{-3}$*, e.g. measuring the residual alkali present after skinning fruit with potassium hydroxide.*

Students will be expected to have carried out simple acid-base titrations, but they may or may not have done this to measure the residual alkali present after skinning fruit.

In the unit test students could be given experimental data for this or any other acid-base titration, and be expected to analyse and evaluate this data.

Examples in unit content

Where 'e.g.' follows a concept students are not expected to have been taught the particular example given in the specification. They should be able to illustrate their answer with an example of their choice.

For instance in this unit, *2.10.2f Halogenoalkanes*, the specification states:

discuss the uses of halogenoalkanes, e.g. as fire retardants and modern refrigerants.

Students will be expected to discuss the use of halogenoalkanes, but they may or may not have looked at their use as fire retardants or refrigerants.

In the unit test students could be asked to discuss some of the uses of halogenoalkanes. This could be those listed as examples or other uses.